Solar power

The sun can be used to generate electricity in 2 ways:

- Using its heat as a heat source
- Utilizing its light in a solar cell



- ‘The sustainable city’ in Dubai
- Big contribution to climate change

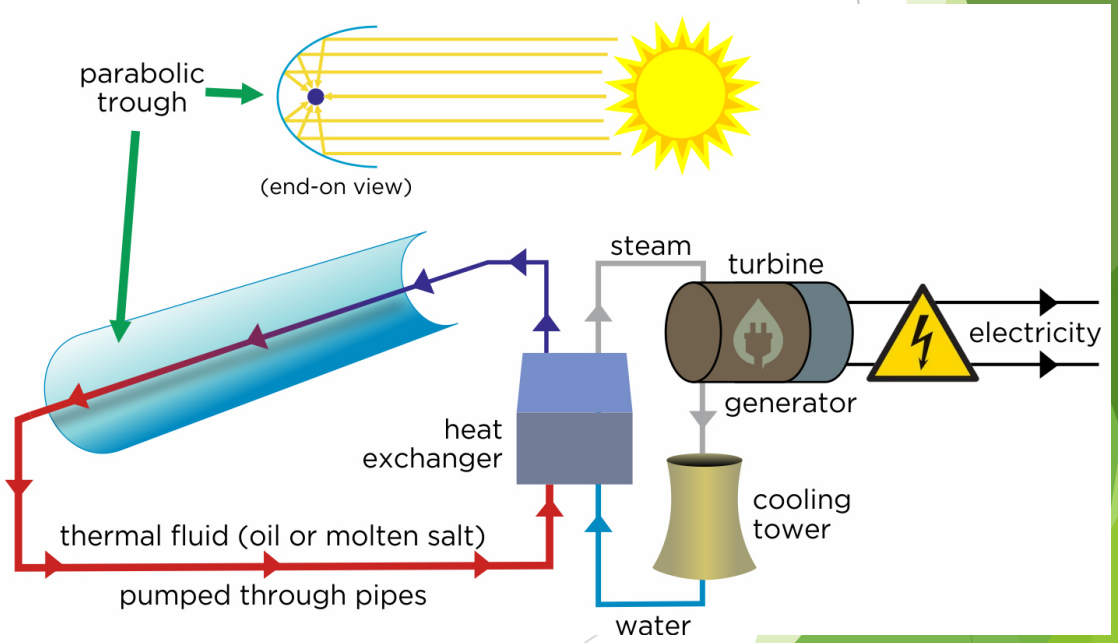
Operations of both ways

- Solar thermal power plants, how do they work?

One type of solar thermal power plant:



Solar thermal power plant operation:

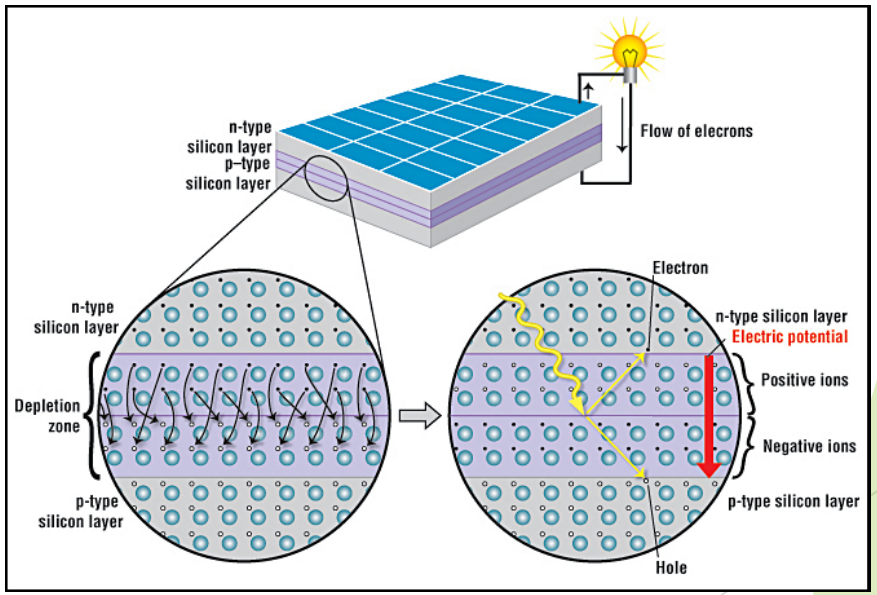


- Solar panels, how do they work?

Solar panels:



Operation of solar panels/cells:



Advantages

The combination of refrigeration systems and solar photovoltaic (PV) technology

System of solar panel powering the freezer:

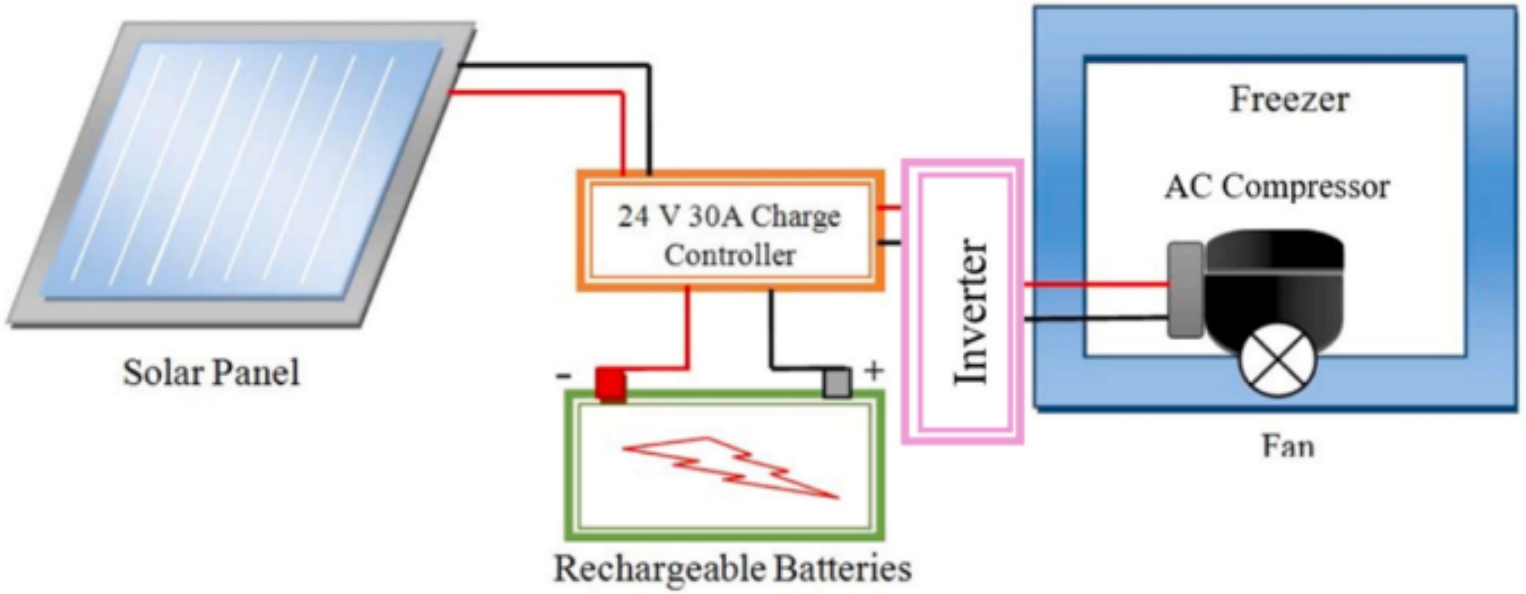


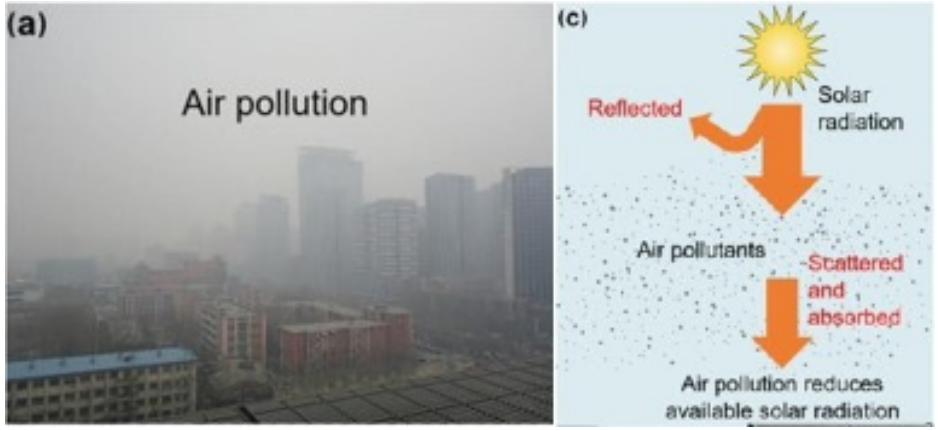
Fig. 3. Schematic diagram of a system with AC compressor (Iqbal et al., 2018).

Limitations

- Both solar power generation systems at night







- Effect of air pollution on solar power generation

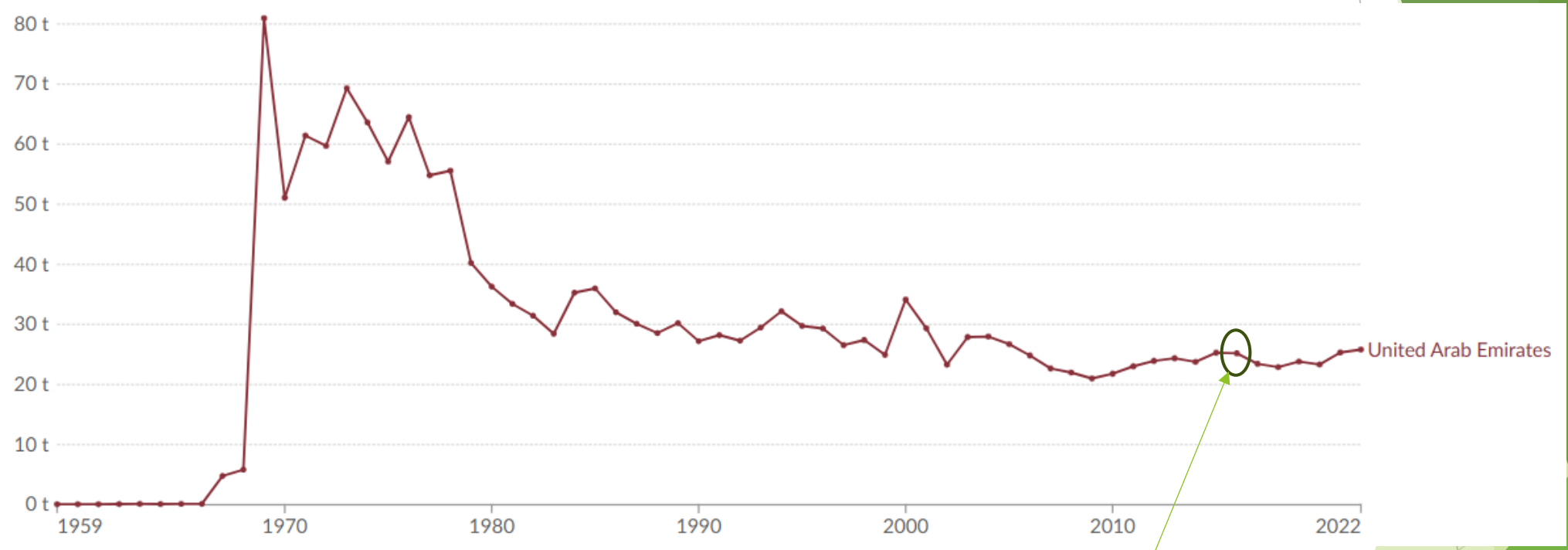
Contribution to energy needs and climate change

Tables showing solar energy capacity increases over time (After installing more solar power generation systems):

Table 3. Installed solar energy capacity in SIDS (in MW) in 2014–2023

Country	2014	2017	2020	2023
Asia-Pacific				
Cook Islands	2	3	6	6
Fiji	2	8	8	11
Kiribati	1	3	3	3

Graph of Per capita Carbon dioxide (CO_2) emissions against year in UAE:



'The Sustainable City' in Dubai was built in 2016

2016

End of presentation

Thank you for your time